# CommonsLens: Automatic Category Suggestions for Wikimedia Commons

Final Report

TEAM-89

Hackathon Problem Statement 3

## 1. Problem and Approach

Categories are how files on Wikimedia Commons are found, but many uploads arrive with few or none, and editors must choose from a huge, growing, long-tailed category graph. A fixed classifier fits poorly: the label space changes daily, most categories have a handful of examples, labels are incomplete (a missing category is not a negative), and editors need to know *why* a category is suggested and *when* the system is unsure. CommonsLens treats categories as retrievable objects in a shared image–text space: a frozen vision–language encoder, several complementary scorers, calibrated fusion, hierarchy-aware decoding, and conformal suggestion sets, with evidence shown for every suggestion and an editor always in the loop. Nothing is applied automatically.

## 2. What Makes the Difference

- **Open vocabulary by design (M1).** Categories are *retrieved*, not classified: a category is suggestible the moment it is named, and new images or editor decisions count immediately, with no retraining.
- **Many weak signals, one calibrated score (M1).** Five complementary scorers are fused on a few hundred candidates, never on all categories: +53% F1@5 over the best single scorer.
- **Missing labels are not punished (M1).** Unlabelled parents of true categories are left out of fusion training instead of counted as negatives, the head uses an asymmetric loss, and decoding is hierarchy-aware; hierarchy-aware precision is reported next to strict precision.
- **Honest about uncertainty (M1, built for the final).** An *Unsure* flag from the content type, and conformal suggestion sets that send an upload to manual categorisation instead of guessing.
- **Explainable (M1).** Every suggestion shows the indexed images that voted for it and the signals behind its score, in line with Wikimedia's transparency principle.
- **Evaluation that matches deployment (M2).** Temporal split, de-duplication and a vocabulary built only from earlier data cut the validation–test gap from .135 to .014; balanced and uniformly random samples allow a bias audit.
- **Robust to real uploads (M2).** Fusion is trained on copies without titles and with degraded images: heavy blur costs 5% F1@5 with a title.
- **Built with testers (M2, final).** Hands-on testing and public tester reports drove more than ten fixes, from centre weighting to a fifth content type, Living thing.
- **Open and licence-clean.** Frozen open-weight encoder, MIT code, openly licensed training data only; it runs on a laptop.

## 3. System

Figure 1 shows the pipeline; the encoder is frozen CLIP ViT-B/32 (SigLIP is supported).

- **Encoder:** L2-normalised embeddings; uploads add a 0.4-weighted embedding of the central 60% crop, so the main subject outweighs the background.
- **Scorers:** (1) $k$NN voting over indexed images, softmax($s/\tau$), editor rejections as negative votes; (2) text prototypes from prompt ensembles; (3) *smoothed prototypes* (text embedding blended with the image centroid, $w{=}n/(n{+}5)$; parent/sibling term tested, Section 4.2); (4) a text-initialised linear head with asymmetric loss for categories with ≥3 images; (5) metadata: title/description embedding, IDF word overlap, year agreement.
- **Fusion:** logistic regression over 17 features on the union of each scorer's top 100, fitted on validation plus copies with metadata removed (30%) or images degraded (30%).
- **Hierarchy decoding:** an ancestor is pushed down when a descendant is confident; the strength is tuned on validation.
- **Conformal suggestion sets:** split conformal calibration on validation, separately for uploads with and without metadata. Members are marked ✓; an empty set means "manual categorisation recommended". $\alpha$ is the smallest value whose median set stays ≤10.
- **Content type first:** Object / Living thing / Scene / Event / Abstract concept, from a classifier on frozen embeddings (939 openly licensed Commons photos, equal share per source category) blended with a vote of the suggestions; "Unsure" below 50%, two types shown when close.
- **Filters:** maintenance and contest categories are never suggested. The display hides bare country names, and over-specific categories (dates, people counts, non-Latin script) unless their score is at least 70%; evaluation scores the model before these display filters.
- **Continual learning:** new categories are suggestible immediately (text prototype, word match), new images are appended to the index, and editor accept/reject feeds back without retraining.

## 4. Data and Protocol

6,303 files from the MediaWiki API: 155 seed categories (107 topical; 48 chosen for balance across continents) plus 1,500 uniformly random files. Hidden and maintenance categories are removed. **Temporal split** by upload time (4,429 / 937 / 937), perceptual-hash and embedding (cos ≥ 0.95) de-duplication. **Realistic validation:** the vocabulary is built from training data (plus seeds and parents) only; validation categories are added after fusion is fitted, giving 24,486 categories at test time; the validation–test F1@5 gap is .014, so validation is a faithful guide. 72% of test labels belong to categories with no

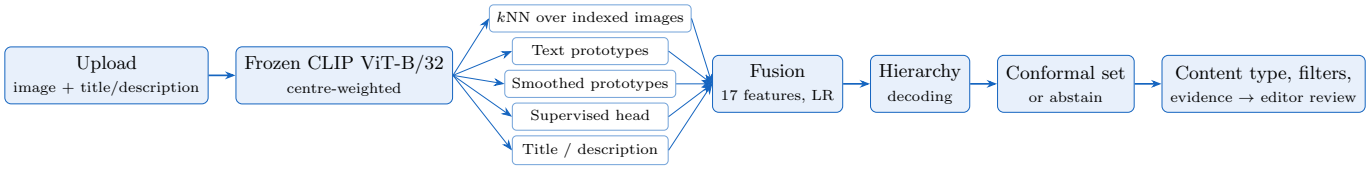



Figure 1: CommonsLens pipeline. Fusion runs only on the union of each scorer's top 100 candidates; editor accept/reject decisions and new categories update the index and vocabulary without retraining.

training image, and only 36% of test labels are in the vocabulary at all, which bounds recall. Encoder: CLIP ViT-B/32; $K$=50, $\tau$ tuned on validation.

## 5. Results

### 5.1 Main results and progression

| **Test** | **image + metadata** | | | | **image** |
|---|---|---|---|---|---|
| | Run 1 | Run 2 | M2 | **Final** | **only** |
| Test images | 190 | 510 | 937 | 937 | 937 |
| P@1 | .284 | .339 | .350 | .348 | .188 |
| F1@1 | .148 | .199 | .237 | .234 | .118 |
| F1@5 | .119 | .160 | .153 | .152 | .085 |
| R@10 | .238 | .310 | .330 | .336 | .193 |
| Hierarchy-aware P@5 | – | – | – | .234 | .121 |
| mAP (macro) | – | – | – | .121 | – |

Run 1→2 added images per category; Run 2→M2 made validation realistic and added balanced and random samples (a harder, more honest test); the final system adds conformal sets, the trained content type and smoothed prototypes. Accuracy holds at the M2 level with these additions. About one in three uploads with a title gets a correct category in first place, and 64% of in-vocabulary uploads get one in the top 10. Hierarchy-aware P@5 (.234) is double strict P@5 (.116): many "wrong" suggestions are a parent or child of a true category, which an editor would often accept. The tuned hierarchy decoding leaves F1@5 unchanged (.152 with and without).

| **Ablation (+meta)** | | | **Recall@5 by frequency** | | |
|---|---|---|---|---|---|
| Ranking | F1@5 | R@10 | Bucket | labels | R |
| Full fusion | **.152** | **.336** | unseen | 2412 | .003 |
| w/o smoothed | .152 | .330 | tail 1–4 | 318 | .200 |
| Lexical only | .099 | .202 | torso 5–19 | 217 | .439 |
| Smoothed only | .083 | .191 | head 20+ | 380 | .821 |
| Head only | .070 | .172 | | | |
| $k$NN only | .069 | .158 | | | |
| Meta text only | .052 | .170 | | | |
| Text proto only | .005 | .020 | | | |

| **By category type** | Object | Living | Scene | Event | Abstract |
|---|---|---|---|---|---|
| Test labels | 2151 | 224 | 597 | 106 | 249 |
| Recall@5 | .123 | .384 | .238 | .217 | .112 |
| Precision@5 | .120 | .184 | .120 | .064 | .059 |

Living things and scenes are easiest; abstract concepts are hardest. Resolution barely matters (F1@5 .155 for ≥2000 px, .143 for 800–2000 px, .144 below 800 px).

### 5.2 Smoothed prototypes (M1's core idea)

| **Validation, prototype only** | | | **Test, R@10 by bucket** | | |
|---|---|---|---|---|---|
| Setting | R@10 | tail | Bucket | text | smoothed |
| Text only | .061 | .061 | all | .020 | **.191** |
| Text + graph | .049 | .053 | tail | **.050** | .025 |
| **Text + images** | **.223** | .061 | torso | .055 | **.240** |
| + graph ($\gamma$=.3) | .221 | .045 | head | .053 | **.797** |
| + graph ($\gamma$=.6) | .213 | .030 | | | |

Image centroids make the prototype scorer nearly ten times stronger on its own (.020→.191), but the full system does not change because $k$NN and the head already carry the same evidence. The parent/sibling term, M1's hypothesis for the tail, *lowered* tail recall on validation, so it was not selected: siblings in the Commons graph often look nothing alike. A first version that *replaced* the text prototypes broke title matching, which compares text with text (metadata-text-only F1@5 .052→.007), so the smoothed set is now a separate signal.

### 5.3 Conformal suggestion sets

| **With metadata** | target | cover. | mean | median | abstain |
|---|---|---|---|---|---|
| $\alpha$=0.3 | .70 | .624 | 13.0 | 12 | .009 |
| $\alpha$=0.4 | .60 | .518 | 4.1 | 3 | .069 |
| $\alpha$=0.5 (default) | .50 | .425 | 2.1 | 2 | .197 |
| *Image only,* $\alpha$=0.5 | .50 | .370 | 9.6 | 9 | .001 |
| Fixed top-$k$: top-3 .484 | top-5 .556 | top-10 .638 | | | |

Coverage counts images with a correct category in the set (in-vocabulary images, 98% of the test set). Sets adapt to difficulty: at $\alpha$=0.4 the median set has 3 categories, beats a fixed top-3 (.518 vs .484), and 7% of images are sent to manual categorisation. Without a title the same guarantee needs sets about five times larger, so the demo encourages adding one. Coverage falls 7–8 points short of the target: test uploads are later than the calibration uploads and meet a larger vocabulary, which breaks the exchangeability the guarantee assumes. Recalibrating on a rolling window of recent uploads (M1's monitoring plan) is required in production.

### 5.4 Bias audit, robustness, efficiency

| **F1@5 by group** | | | | **Robustness, F1@5** | | |
|---|---|---|---|---|---|---|
| Group | $n$ | +meta | img | Degradation | +meta | img |
| Objects | 318 | .190 | .141 | none | .152 | .085 |
| Scenes | 109 | .239 | .170 | blur $r$=2 | .156 | .077 |
| Events | 20 | .233 | .206 | blur $r$=5 | .145 | .054 |
| Abstract | 127 | .051 | .029 | JPEG q10 | .151 | .074 |
| Europe | 42 | .274 | .037 | down ×.15 | .152 | .063 |
| Africa | 71 | .207 | .006 | | | |
| S. America | 48 | .087 | .009 | | | |
| Asia | 47 | .087 | .034 | | | |
| N. America | 43 | .036 | .012 | | | |

**Efficiency** (laptop, no server GPU): end to end p50

30.1 ms, p95 32.6 ms per image; scoring and ranking 1.0 ms; 5,366 indexed images and 24,486 categories in 58.3 MB of memory (84.3 MB on disk).

### 5.5 Content type

The classifier is trained on 939 photos (Object 175, Living thing 216, Scene 193, Event 176, Abstract 179) and reaches 82.9% on held-out photos across the five types. On a hand-labelled pack of 250 living subjects (88 of them deliberately hard: partly hidden, old or low-contrast photos), 91.6% were typed Living thing and 88.6% of the hard ones got the right broad type. Portraits were the exception (4 of 14), traced to a collection bug that left the people source categories unused; it is fixed and the classifier retrained. Test-pack photos are never used for training.

### 5.6 Newest uploads (deployment distribution)

To see what a deployment would face, we sampled 480 files uploaded 7–30 days before evaluation, from 25 random moments with at most five per uploader and none from the training data. Only **2.0%** of their labels exist in our 24,486-category vocabulary (36% on the seed-based test set), so F1@5 is .002 and R@10 .004, about a fifth of what is reachable. On real uploads accuracy is capped by *vocabulary coverage*, not by the model: a deployment must index the full Commons category list. CommonsLens can add categories without retraining (text prototype and title matching, our strongest single signal), so this is a data step, but a necessary one.

## 6. Findings

- **Fusion is the main win:** .152 vs .099 for the best single scorer (+53%).
- **Metadata is essential for places and robustness:** Africa .207 with a title vs .006 image only; under heavy blur F1@5 drops 5% with metadata but 36% image only.
- **The long tail is the bottleneck:** recall@5 .821 for categories with ≥20 images, .003 for unseen ones, which hold 72% of test labels. Neither text prototypes nor graph smoothing closed this gap, and on the newest uploads only 2% of labels are in the vocabulary.
- **Regional disparity:** with metadata, Europe .274 and Africa .207 but North America .036; image-only accuracy is near zero for every region. Groups are small (42–71 images), so the gaps are indicative only.
- **Evaluation pitfalls we caught:** a validation vocabulary that included validation labels (val–test gap .135→.014 once fixed), and a display-only filter wrongly applied inside the model, which silently zeroed the scores for photos from several regions.

## 7. Testing

77 automated tests (offline end-to-end train → evaluate → predict → update, conformal coverage on synthetic data, smoothing, collectors, demo). Hands-on testing with our own photos and tester reports from the public demo led to: centre weighting (people pulled towards "Marathons"), buildings as Scenes, Living thing as a fifth type, mixed types (a fruit market is Object + Scene), hiding dated and non-Latin categories, filtering "Wiki Loves" contest categories, and de-duplicating repeated reports.

## 8. Responsible Use and Licensing

Only openly licensed Commons files are used for training and indexing, and each keeps its licence. External datasets suggested during development (ObjectNet, WIDER FACE, DTD, COCO) were *not* used for training: their terms forbid training on them, limit use to non-commercial research, or mix such licences, which would conflict with the MIT licence and the sponsor's commercial licence. CommonsLens never identifies people; it only types them as living things. Suggestions are always reviewed by an editor, with the evidence images shown.

## 9. Editor Experience

Each result starts with a ⋆ row giving the content type (or two, e.g. a fruit market is Object + Scene) and an *Unsure* banner when the type is below 50%. Suggestions follow with their score; rows marked ✓ form the conformal set, and notes explain abstention, a missing title or a low-quality image. For every suggestion the demo shows the indexed images that voted for it and the signals behind its score. Editors can accept or reject suggestions and add new categories; testers can report a problem, which is stored with what the system showed. A public read-only mode hides the editing panels.

## 10. Deployment and Scalability

The encoder is frozen, so no GPU training is needed and a new deployment only builds an index. Scoring and ranking take 1 ms per image after encoding; the index is a FAISS flat, HNSW or IVF-PQ index chosen by configuration, and candidates keep fusion independent of the number of categories. On Commons the system would run asynchronously on upload events and store suggestions with their evidence for editors; nothing blocks or changes an upload. New categories, new images and editor decisions take effect immediately; the fusion model and conformal thresholds are refitted periodically on recent uploads.

## 11. Reproducibility

Every number in this report comes from the repository's scripts (`prepare_data.py`, `train.py`, `evaluate.py`, `train_content_type.py`) with a fixed seed; the README lists the commands and links to the built artifacts. All code is MIT-licensed; third-party models, libraries and data are credited in the README.

## 12. Limitations and Future Work

Only 36% of test labels (2% for the newest uploads) are reachable, so the first step is indexing the full Commons category list, then the tail (multilingual category names, geotags, mining sibling categories with editors). Also pending: a SigLIP vs CLIP comparison (supported in the code), a human-reviewed score for "wrong" suggestions that are actually valid, and rolling conformal recalibration.